Digital Logic Fundamentals

PRE-WORK
SEMICONDUCTOR IC DESIGN FOUNDATION
AND READINESS PROGRAMME

2-Hour Training Module

Module 2 at a glance

ORIENTATION

Establish a foundation in how digital systems represent, process and store information. Participants will understand the logic concepts that form the basis of digital circuits and integrated circuits.

module 2. pre test
https://forms.gle/TMGGoNCKG9TF15F16

Learning outcomes

ORIENTATION

Binary and hexadecimal number systems; Boolean logic; logic gates such as AND, OR, NOT, NAND, NOR and XOR; truth tables; combinational and sequential logic; clock signals; registers and memory elements; flip-flops; counters; state machines

Why this is important

CONCEPTS

This knowledge helps participants understand how a digital circuit behaves before it is translated into an actual chip design

Number Systems Concept

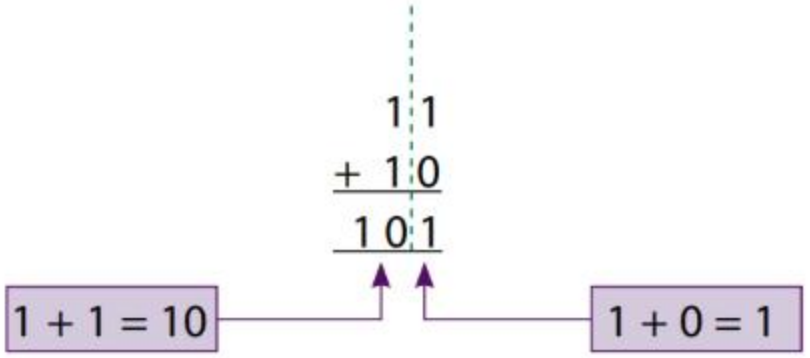
Number system	Base or radix (b)	Symbols used (d_i or d_{-f})
Binary	2	0, 1
Octal	8	0, 1, 2, 3, 4, 5, 6, 7
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9
Hexadecimal	16	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, *A*, *B*, *C*, *D*, *E*, *F*

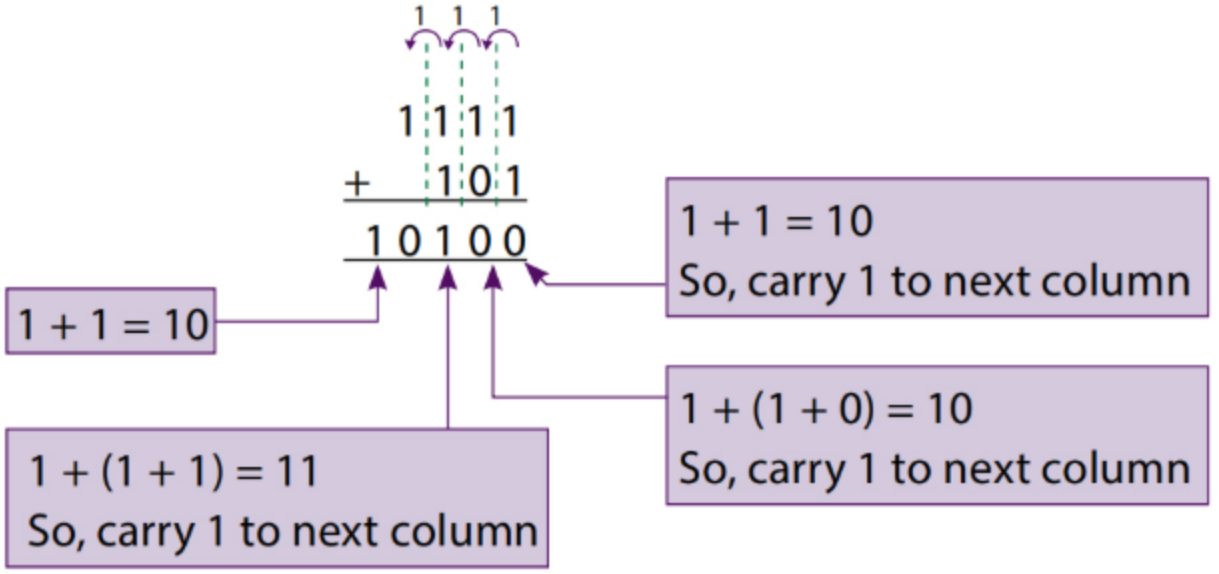
Quick check: What will be the number after 9 for Decimal system? 0 . For Octal, after 7-->0. Binary number is popular number system in digital system.

Binary and Decimal number

Binary number				Decimal number	
B_3	B_2	B_1	B_0	D_1	D_0
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	0	2
0	0	1	1	0	3
0	1	0	0	0	4
0	1	0	1	0	5
0	1	1	0	0	6
0	1	1	1	0	7
1	0	0	0	0	8
1	0	0	1	0	9
1	0	1	0	1	0
1	0	1	1	1	1

Arithmetic


1 1
+ 1 0
1 0 1
1 + 1 = 10
1 + 0 = 1


1 1 1
1 1 1 1
+ 1 0 1
1 0 1 0 0
1 + 1 = 10
1 + 1 = 10
So, carry 1 to next column
1 + (1 + 1) = 11
So, carry 1 to next column
1 + (1 + 0) = 10
So, carry 1 to next column

Boolean

Boolean algebra is mathematics of logic that have only two possible values : 0 or 1; False or True.

Application with the advent of digital electronics.

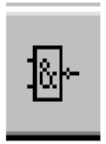
- **Logical Expression Simplification.**
- **Arithmetic Circuits:** Helps design binary adders, subtractors, multipliers, and dividers.
- **Memory Design:** Used in flip-flops, latches, and registers for data storage and state control.
- **Combinational & Sequential Circuit Design:** Used to design and optimize circuits like multiplexers, decoders, and counters.
- **Error Detection & Correction**

Basic Gates

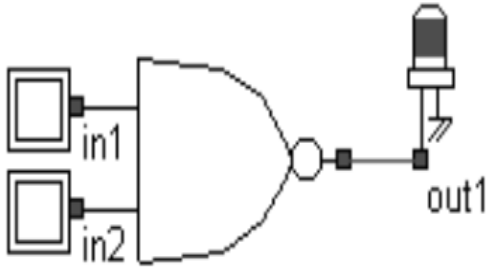
Name	Logic symbol	Logic equation
INVERTER		Out=~in;
AND		Out=a&b;
NAND		Out=~(a.b);
OR		Out=(a|b);
NOR		Out=~(a|b);
XOR		Out=a^b;
XNOR		Out=~(a^b);

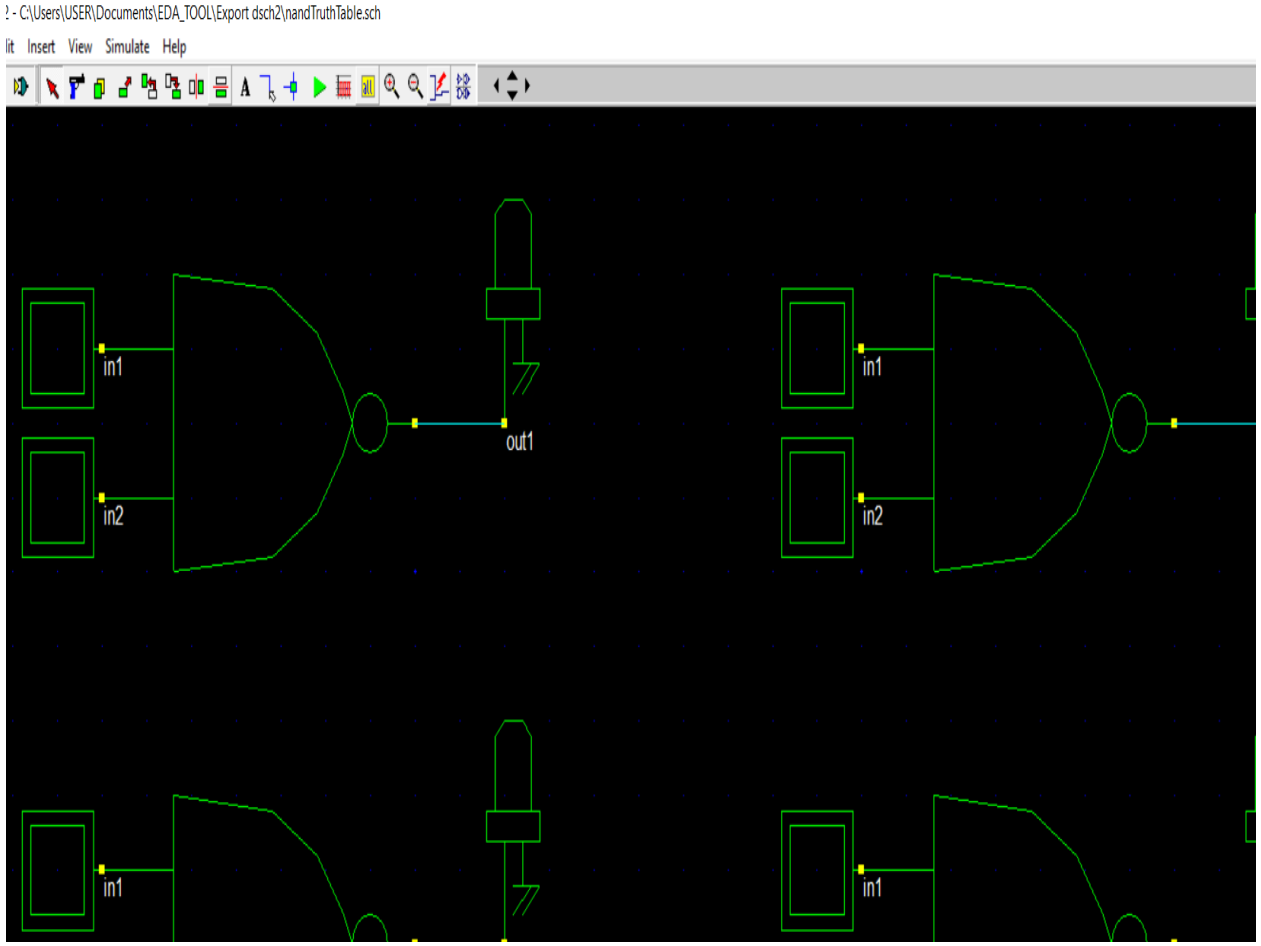
Quick check: Which logic gate corresponds to addition in Boolean algebra? What about logic gate AND? -Multiplication

Truth Table and Symbol of NAND gate



in1	in2	Out
0	0	1
0	1	1
1	0	1
1	1	0



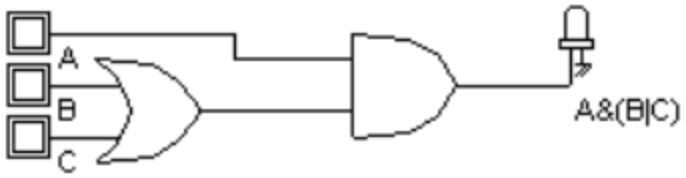


What is the Output Boolean expression ?

Combinational or Complex gate

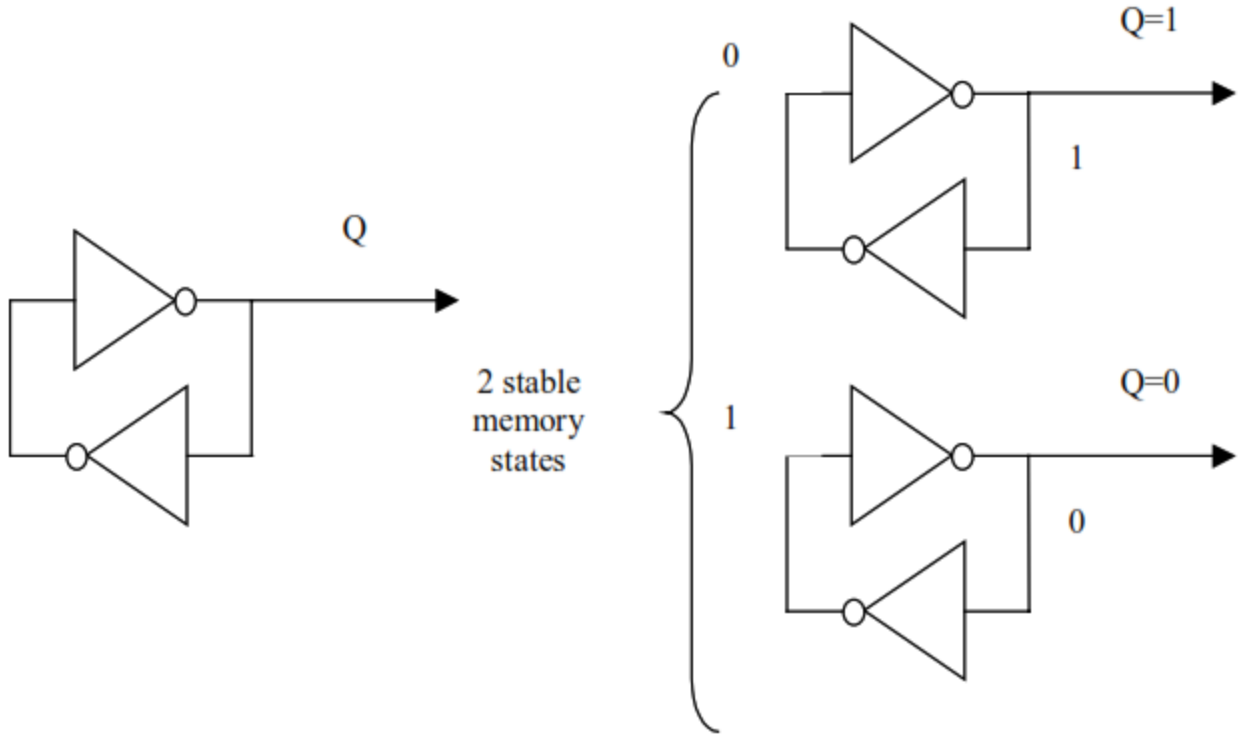
```
F= A&(B|C)
```

Bit operation	Verilog symbol used in complex gates
Not	~
And	&
Or	|



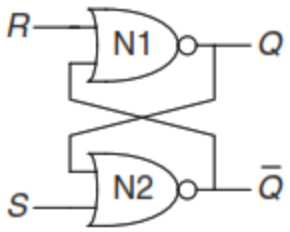
The circuit is built using a 2-input NOR and a 2-input AND cell,

Basic Latch or Memory cell



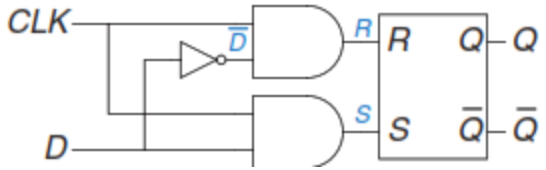
Quick check: The memory cell is volatile cell. The memory value remain if the power supply is still applied.

RS Latch -> D Latch->D Flip Flop



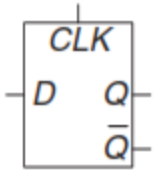
S	R	Q	$\bar{Q}$
0	0	Q_{prev}	$\bar{Q}_{prev}$
0	1	0	1
1	0	1	0
1	1	0	0

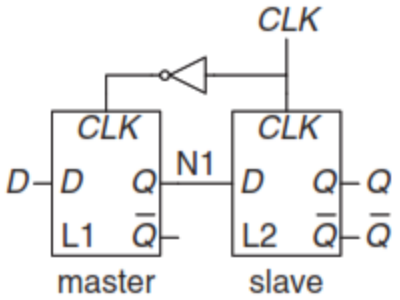
RS Latch



CLK	D	$\bar{D}$	S	R	Q	$\bar{Q}$
0	X	$\bar{X}$	0	0	Q_{prev}	$\bar{Q}_{prev}$
1	0	1	0	1	0	1
1	1	0	1	0	1	0

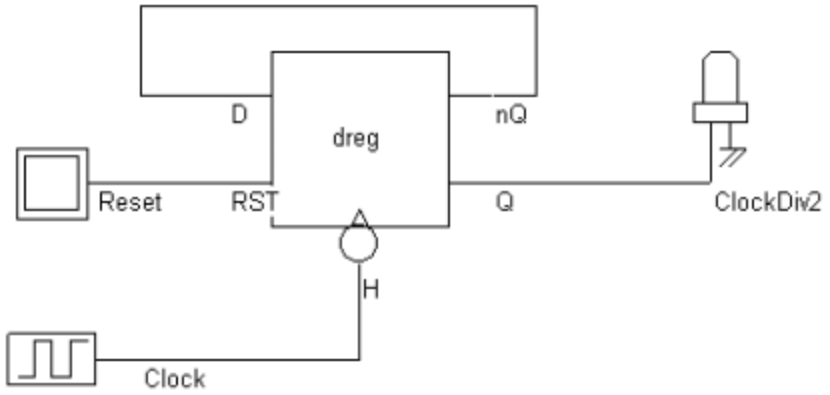
D Latch

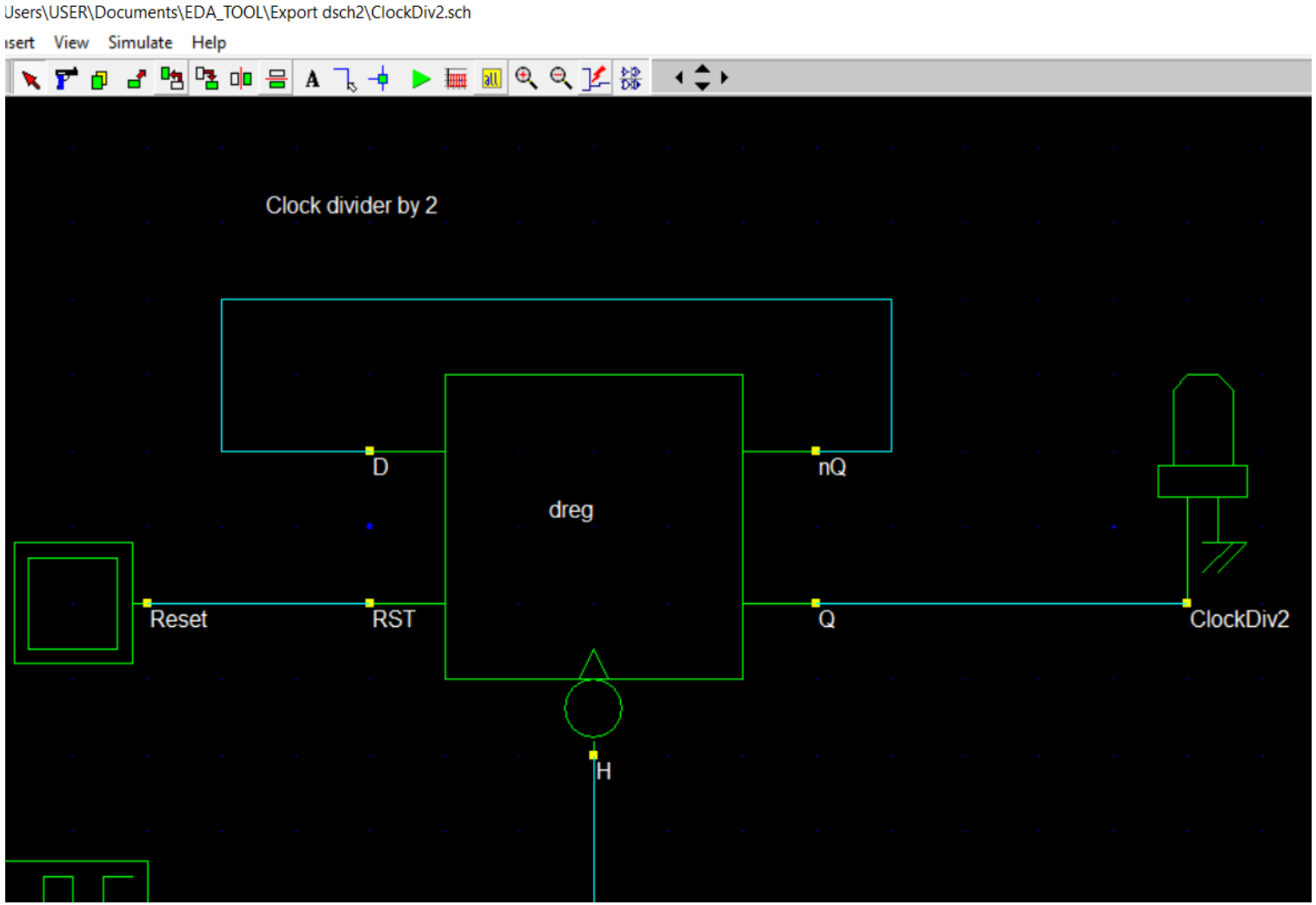




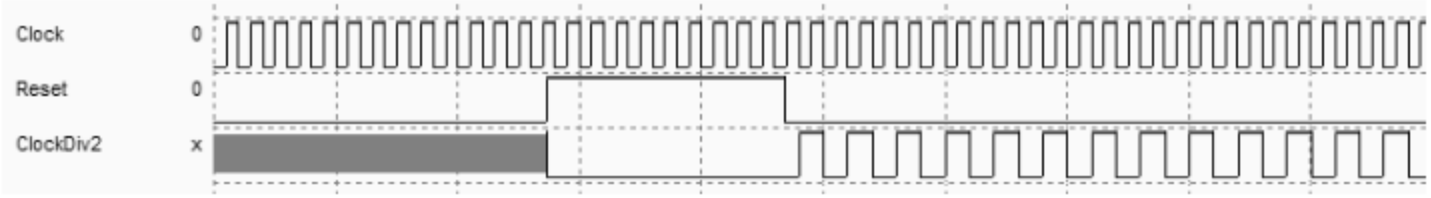
D Flip Flop

Counter

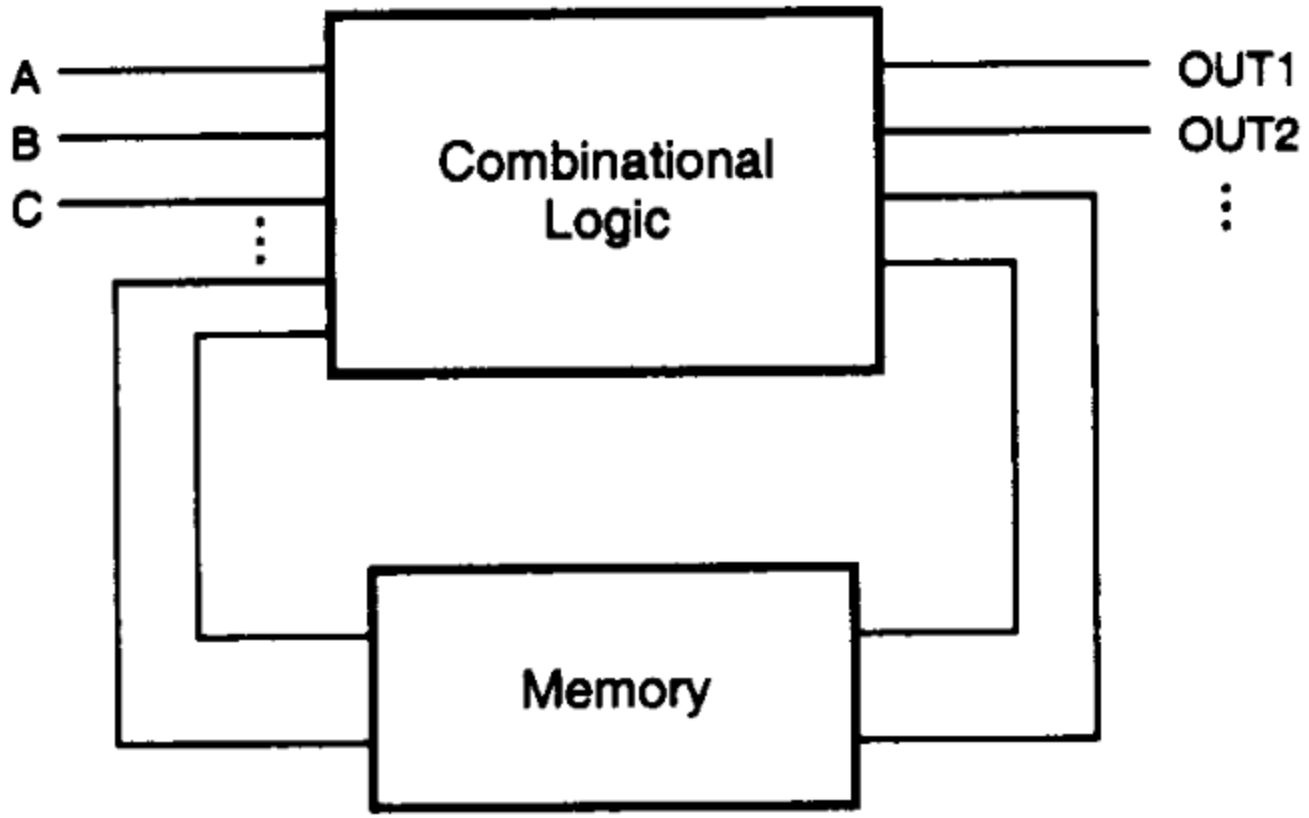




The one-bit counter is able to produce a signal featuring half the frequency of a clock.

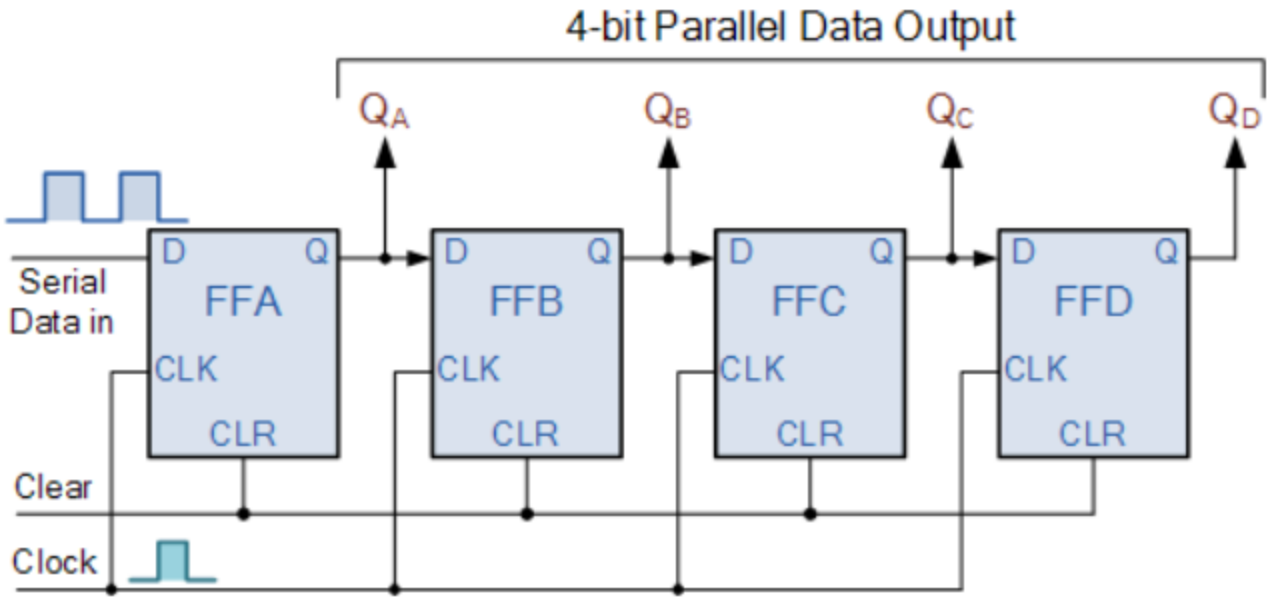


Sequential Circuit

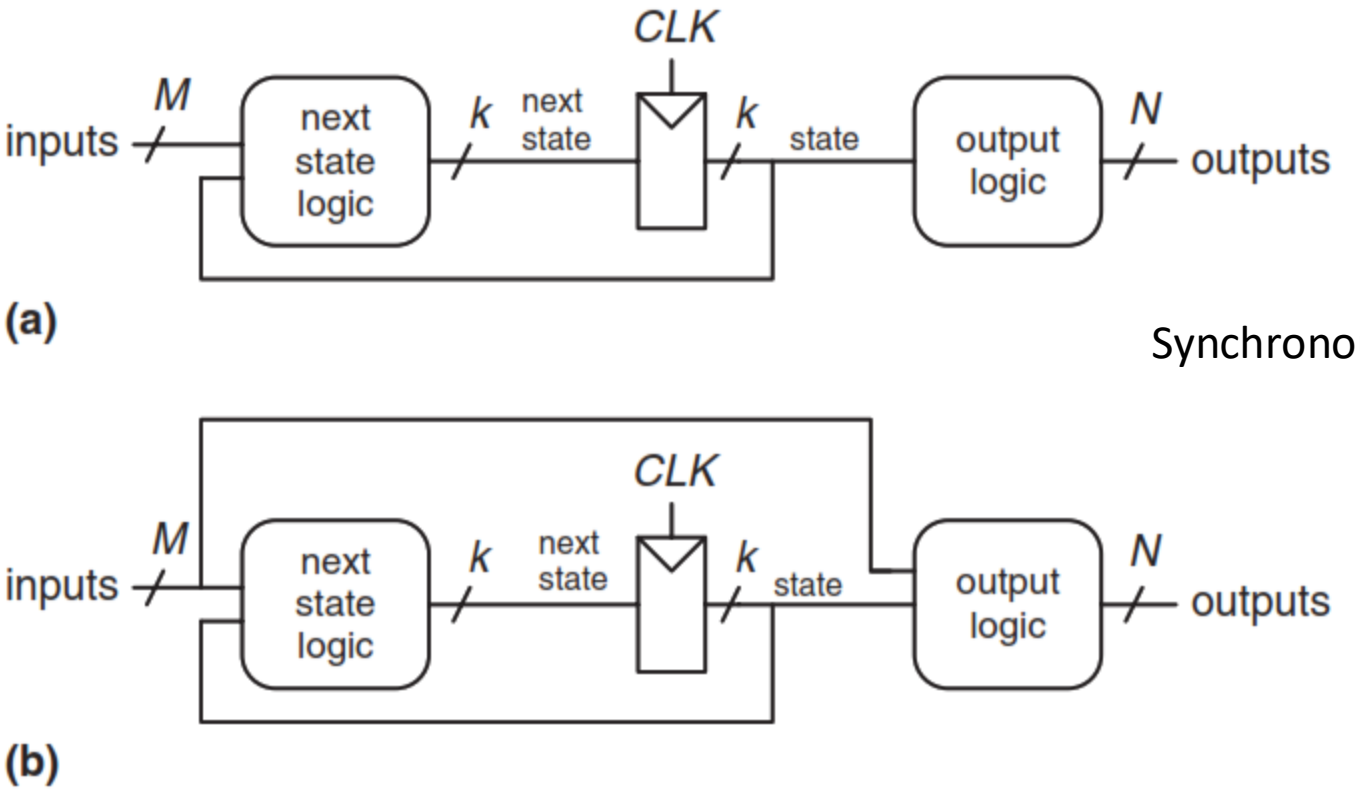


Register

Registers are common sequential elements that remember their previous input. Series of FF.



FSM



Synchronous sequential circuits.

Finite state machines: (a) Moore machine (b) Mealy machine

Knowledge check

WRAP-UP

1. What is the difference between binary and decimal number systems?
2. The importance of Boolean.
3. Clock and sequential are related.

Module 2 post test
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